Chapter 1 Vector Bundles

1.1 Definitions,Constructions

Definition 1.1.0.1 An n-dimensional vector bundle is a map $p: E \to B$ together with

- a real vector space structure on $p^{-1}(b)$(**fibre**) for each $b \in B$;
- There exists a covering of B by U_α such that: Equipped with homeomorphism

$$h_\alpha : p^{-1}(U_\alpha) \stackrel{\cong}{\to} U_\alpha \times \mathbb{R}^n$$

which is a homemorphism. And $h_\alpha | p^{-1}(b) : p^{-1}(b) \to \{b\} \times K^n$ is a vector isomrophism.
such h_α is called **trivialization** of the vector bundle.

In principle, $K = \mathbb{C}, \mathbb{R}$.
♣ **Example 1.1.0.2** ...
♣ **Example 1.1.0.3** ...
♣ **Example 1.1.0.4** (Tangent Bundle)
♣ **Example 1.1.0.5** (Normal Bundle)
♣ **Example 1.1.0.6** ...

1.2 Classifying Vector Bundles

1.2.1 Pullback Bundles

1.2.2 Clutching Functions

1.2.3 Cell Structures on Grassmannians

Chapter 2 K-Theory